

LAB 5 – Telemetry Stream Audit

Submission Documentation

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1. Short Note: Shutdown Flag and Outer Loop Else

The variable `shutdown` works as a flag that records whether a "STOP" signal has appeared anywhere in the telemetry stream. At the beginning of the program, `shutdown = False`, meaning that no shutdown has been detected. While the inner loop checks each reading, encountering "STOP" changes the flag to `True` and immediately stops the current batch using `break`. After the inner loop, the outer loop checks the flag. If it is `True`, another `break` stops the outer loop. Therefore, the outer loop's `else` clause runs only when the outer loop finishes normally without executing its `break`. Since the only condition that sets `shutdown` to `True` is the detection of "STOP", the message "Audit complete: No system-wide failures" is printed only when no "STOP" signal was ever seen. This makes the flag a simple and reliable way to distinguish a normal audit from an emergency shutdown.

2. Python Code

```
telemetry_stream = [
    [22.5,23.0,22.8],
    [25.1,"ERR",24,9],
    [30.2,35.5,40.1],
    [22.0,22.1,"STOP"],
]

shutdown = False

for i in range(len(telemetry_stream)):
    batch = telemetry_stream[i]
    print(f"---Auditing batch {i} : {batch}---")

    previous_value = None

    for reading in batch:
        if reading == "STOP":
            print(f"Emergency Shutdown at Batch {batch}:")
            shutdown = True
            break

        if reading == "ERR":
            print(f"Noise ignored at {i} ,(ERR)")
            continue

        if isinstance(reading ,(int,float)):
            if reading > 35.0:
                print("Anomaly Detected at Batch",i,":",reading)
```

```

        if previous_value is not None:
            delta = abs(reading - previous_value)

            if delta > 5.0:
                print(
                    f"Spike detected at {i} : "
                    f"{previous_value} -> {reading} "
                    f"Delta{delta:.1f}"
                )

            previous_value = reading

    if shutdown:
        break

else:
    print("Audit complete: No sysytem-wide failures")

```

3. Program Output

```

---Auditing batch 0 : [22.5, 23.0, 22.8]---
---Auditing batch 1 : [25.1, 'ERR', 24, 9]---
Noise ignored at 1 ,(ERR)
---Auditing batch 2 : [30.2, 35.5, 40.1]---
Anomaly Detected at Batch 2 : 35.5
Spike detected at 2 :30.2 -> 35.5 Delta5.3
Anomaly Detected at Batch 2 : 40.1
---Auditing batch 3 : [22.0, 22.1, 'STOP']---
Emergency Shutdown at Batch [22.0, 22.1, 'STOP']:

```

4. Conclusion

The program successfully detects errors, anomalies, spikes, and the "STOP" signal. When "STOP" is found, the shutdown flag becomes `True`, causing both loops to terminate. As a result, the outer loop's `else` clause does not execute.